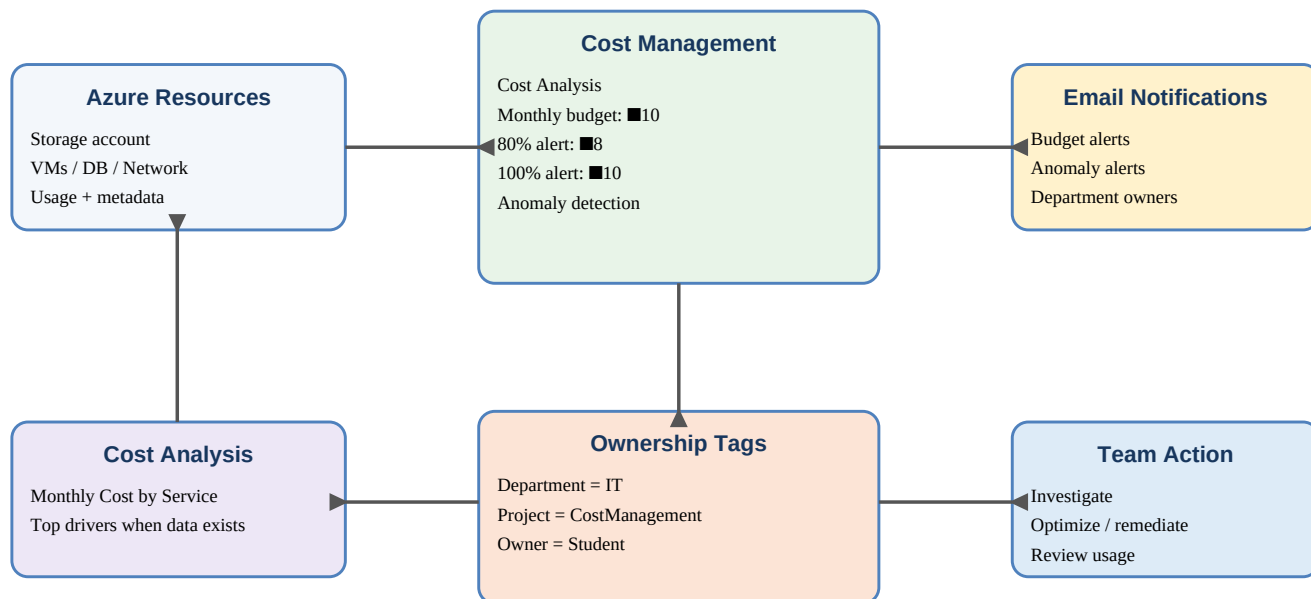


Architecture — Azure Cost Management, Budget & Anomaly Alerting

Azure Cost Monitoring and Alerting Architecture



Architecture overview

This setup uses Azure Cost Management as the control layer for an Azure for Students subscription. Azure resources generate usage and cost data. Resource tags provide ownership metadata, Cost Analysis provides the reporting view, a monthly budget evaluates actual spend against configured thresholds, and an anomaly rule identifies unusual daily spending patterns and sends email notifications.

Starting with the Azure resources

The live implementation uses the resource group **RG-CostManagement** and storage account **costalertstorage001**. The storage resource is tagged with **Department=IT**, **Project=CostManagement**, and **Owner=Student**.

The cost-control layer

At the subscription scope **Azure for Students**, the project uses a monthly budget of **█10**. An actual-cost alert is configured at **80% (█8)** and another at **100% (█10)**, with email notifications configured.

Description

What this architecture is actually showing

The workflow connects resource usage, ownership metadata, cost analysis, budget thresholds, anomaly detection, and human action. The purpose is to make spending visible and produce early signals that an owner can investigate.

1. Resource usage generates cost

Azure resources consume services and generate usage records. Cost Management uses subscription cost data for analysis. The student implementation avoids unnecessary paid resources because student credits are limited.

2. Tags provide ownership

Consistent tags connect resources with a department, project, and owner. The implemented values are **Department=IT**, **Project=CostManagement**, and **Owner=Student**.

3. Budget thresholds provide early signals

The monthly **■10** budget evaluates actual cost. At **■8**, the 80% warning threshold is reached; at **■10**, the 100% threshold is reached. Email notifications are configured.

4. Cost Analysis identifies cost drivers

The saved view is named **Monthly Cost by Service**, uses the Azure for Students subscription, and groups spending by service name. The current subscription has not reported billable cost data, so an actual top-five cost-driver list cannot yet be produced. The configured view is ready when data becomes available.

5. Anomaly detection highlights unusual changes

The active rule **Daily anomaly by resource group** is configured at the Azure for Students subscription scope. It flags unusual cost behavior and sends email notifications so the team can investigate the affected resource group and service.

Why this architecture is useful

- Threshold alerts provide an early warning before the configured monthly budget is fully reached.
- Tags make resources easier to associate with departments, projects, and owners.
- Cost Analysis provides a repeatable monthly method for identifying cost drivers once cost data is available.
- Anomaly detection turns unusual spending into an investigation item.

Bottlenecks students may encounter

- Cost Analysis may initially show no data because a new subscription has not yet reported billable usage.
- Budget alerts are notifications; they do not automatically stop resources.
- Missing or inconsistent tags make ownership and cost allocation harder.
- Student credits are limited, so testing should avoid unnecessary paid resources.

Live evidence: Resource + tags, **■10** budget with 80%/100% alerts, Monthly Cost by Service view, and Active Daily anomaly by resource group rule.